
PH 501: PH 501

Course Code	Solid State Physics
Course Name	PH 501
L-T-P-C	3-0-0-3
Approval	2nd Adhoc Senate (May 2012), in the Minor in Applied Physics curriculum

Course Contents

- **Module I**
- Crystal structure, Brillouin zone [2 Lectures]
- **Module II**
- Quantum mechanics of electrons in a solid: Electronic degrees of freedom: from a single atom to N atoms, "Free" electron description- why should it ever work?, "Independent" electron description existence of a Fermi surface, Electron' bands: metal, semiconductor and insulator, Quantum well, dot, wire, nanotube. [6 Lectures]
- **Electron Transport**
- Electrons in a field, Boltzmann transport, Quantum of conductance, Meaning of Ohm's law, coherent transport, From atoms to quantum devices. [6 Lectures]
- **Phonons**
- Vibrations of crystals with monoatomic basis, Two atoms per primitive basis Quantisation of elastic waves, Phonon Momentum [3 Lectures]
- **Magnetism**
- A macroscopic quantum phenomenon, Magnetic coupling of electrons: Para, Ferro and Diamagnets, Curie's Law, Pauli Paramagnetism, Curie-Weiss theory, No classical analogue: Bohr van Leueen theorem, Magnetic interactions: long range order, magnetic excitations, Spintronics applications: using itinerant electron spin for transport a new paradigm, new electronic materials, GMR and CMR [10 Lectures]
- **Superconductivity**
- Basic phenomena, Meissner effect, London equation, Towards a pairing mechanism: Cooper problem, BCS theory, experimental verification, Type II superconductors [10 Lectures] .
- **Module VII**
- Two dimensional electron gas in a FET, IQHE: MOSFET configuration: 2D electron confinement, Electrons in a magnetic field: Landau levels, Hall effect: the quantized version. [5 Lectures]

Note: Experimental techniques associated with each chapter will also be covered

Text Books

1. Ashcroft & Mermin, **Solid State Physics**, Cengage learning.
2. G. D. Mahan, **Condensed Matter Physics in a Nutshell**, Princeton University Press.
3. Charles Kittel, **Quantum Theory of Solids**, Wiley.

References

1. A. H. MacDonald, **Quantum Hall effect**, Kluwer Academic.
2. B. Martin Janssen, Janos Hajdu, **Introduction to the theory of the integer quantum Hall Effect**, VCH.
3. S.M. Sze and kwok K.Nag, **Physics of Semiconductor Devices**.